

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A physicist measures the velocity v of a particle as a function of time t and finds that $v = 2t + 1$. Find the inverse of this function and interpret its meaning in the context of the experiment. Show all work and explain your answer.

Q10. A spaceship's altitude h above the Earth's surface is given by $h = 5t^2 - 2t + 1$, where t is time in minutes. Find the inverse of this function and use it to determine the time at which the spaceship reaches an altitude of 10 km. Show all work and explain your answer.

Chef's Tips

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- Tip 1: Ensure students understand the concept of inverse functions and how to find them algebraically.
- Tip 2: Emphasize the importance of verifying inverses using composition.
- Tip 3: Provide real-world contexts for students to interpret inverses and make connections to the physical world.